Copernicus Global Land Operations "Cryosphere and Water"

" CGLOPS-2"

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PRODUCT USER MANUAL

LAKE AND RIVER WATER LEVEL

300M

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List of Acronyms

C-GLOPS	Copernicus Global Land Operations
ECV	Essential Climate Variable
FTP	File Transfer Protocol
GDR	Geophysical Data Records
HR	High Rate
HYSOPE	**Hy**drologie **S**patiale **Opé**rationnelle
IGDR	Interim Geophysical Data Record
J2	Jason-2
J3	Jason-3
NRT	Near Real Time
NTC	Non Time Critical
PDF	Probability Distribution Function
PUM	Product User Manual
QAR	Quality Assessment Report
S3A	Sentinel-3A
S3B	Sentinel-3B
S6A	Sentinel-6A
STC	Short Time Critical
VS	Virtual Station
WGS84	World Geodetic System 1984
WSH	Water Surface Height

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1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Copernicus *Global* Land Service is a component of the Copernicus Land service and is operational since January 2013, providing a series of bio-geophysical products describing the status and the evolution of the land surface at global scale. Production and delivery of the parameters are to take place in a timely manner and are complemented by the constitution of long-term time series. The Service consists of i) a production component providing the bio-physical variables; ii) an algorithm maintenance and evolution component implementing the adjustments needed to the processing chains; iii) a data distribution and user interface activity managing the website and the dissemination of products over the web (FTP) and through EUMETCAST.

Among the “Cryosphere and Water” thematic area of the Copernicus Global Land Service, **this document describes the Water Level (ECV T.1.2) products as observed over lakes and rivers from satellite altimeters** such as Jason-3 -and then Sentinel-6A since April 2022-, Sentinel-3A and Sentinel-3B for near real-time products. Only open water rivers and lakes are considered now, as the radar return echo is contaminated by vegetation.

The water level products thus aim at providing users with worldwide historical and near-real-time water level time series on inland water targets located beneath satellites tracks. Satellite altimetry measurements homogeneously cover all continents between the maximum latitudes reached by each satellite (66° for Jason/Sentinel-6A and 81° for Sentinel-3) and are currently only being possible at nadir. Hence the ground coverage is that of the nadir of the altimeter constellation used. Future missions with wide-swath altimetry technology, such as SWOT - launched in December 2022 -, will allow increasing the spatial coverage of inland waters.

The lake and river water level products are generated daily, within 1.5 days after data acquisition by the satellites. The temporal repeatability for each product is also strongly dependant on the orbital characteristics of the altimeters: for river products, the stations are covered once per satellite cycle (10 days for Jason/Sentinel-6 and 27 days for the Sentinel-3). As for the lakes, the larger their surface allows them to be overflown by several satellite passes, the smaller is the time between two revisits.

This Product User Manual (PUM) is a self-contained document which provides the necessary information on how to use the Water Level products in an efficient and reliable way.

1.2 SCOPE AND OBJECTIVES

The Product User Manual (PUM) is the primary document that users should read before handling the products. It gives an overview of the product characteristics, in terms of algorithm, technical characteristics, and main validation results.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the users' requirements, and the expected performance
- Chapter 3 summarizes the retrieval methodology
- Chapter 4 describes the technical properties of the product
- Chapter 5 summarizes the results of the quality assessment

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Call for tenders JRC/2020/OP/3213-Framework Service Contract – Operations of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications of 17th December 2020

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service.
CGL-TS-002_user_requirements	User Requirements of the Global Component of the Copernicus Land Service "Cryosphere and Water"
CGLOPS2_ATBD_LakeAndRiverWaterLevel300m-V2.2	Algorithm Theoretical Basis Document for 300m Lake and River Water Level Version 2.2 product
CGLOPS2_QAR_LakeAndRiverWaterLevel300m-V2.2	Report describing the results of the scientific quality assessment of the 300m Lake and River Water Level Version 2.2 product

1.4.3 Output

Document ID	Descriptor
CGLOPS2_PUM_LakeAndRiverWaterLevel300m-V2.2	Product User Manual summarizing all information about the 300m Lake and River Water Level Version 2.2 product

2 REVIEW OF USERS REQUIREMENTS

- **Definition**:
  - The Water Level is the measure of the absolute height of the reflecting water surface beneath the satellite with respect to a vertical datum (geoid) and expressed in cm.

Water Level (Lakes and Rivers)		
Content of the data set (01)		
PR-PS-0101	Content of main data file	▪ The data file shall contain the following information on separate layers: - WL value with associated flags - A measure of the uncertainty
PR-PS-0102	Flags	The product shall contain the following flags: ▪ Current C-GLOPS flags ([RD05]) ▪ Water/no water ▪ Seasonal/permanent water body
PR-PS-0103	Information on synthesis product	Synthesis product shall contain the following per-pixel information: date of acquisition or number of pixels used in compositing, depending on method
Spatio-Temporal Features (02)		
PR-PS-0201	Geographic Coordinates - Projection System	The product shall be distributed in the following projection: ▪ Geographic Lat/Lon - WGS1984 ▪ EASE-Grid 2.0
PR-PS-0202	Spatial coverage	The product shall be distributed globally based on a harmonized identification of lakes and rivers for all the lakes/rivers products ▪ Lakes area: - Target = 1 km x 1 km ▪ Rivers width: - Threshold = 500 m - Target = 100 m
PR-PS-0203	Spatial resolution	Non gridded product, does not apply
PR-PS-0204	Spatial data selection	The portal shall allow selection of ROI by shapefile, coordinates and bounding box
PR-PS-0205	Temporal coverage	Times series of 10 years minimum are required

Water Level (Lakes and Rivers)		
PR-PS-0206	Generation frequency/ synthesis period	▪ A monthly composite of the product shall be distributed ▪ A 10-day composite of the product shall be distributed ▪ A daily product should be distributed (especially for rivers)
PR-PS-0207	Geometry accuracy	1/3 pixel
PR-PS-0208	Coordinate position	The coordinate reference shall be the pixel centre
	Distribution (03)	
PR-PS-0301	Format	The product should be distributed in different convenient formats that are supported by most of the remote sensing and GIS tools: GeoTiff, NetCDF, ENVI, HDF5
PR-PS-0302	Access	▪ Free and open access by FTP to all registered users via the Global Land service portal (http://land.copernicus.eu/global/) ▪ https access
PR-PS-0303	Type of service	NRT and offline
PR-PS-0304	Timeliness	All the near real time (NRT) products shall be disseminated within a period after the end of the synthesis period: ▪ monthly product: 1 week (threshold) ▪ 10-day: 3 days ▪ Daily (target): 1 to 2 hours
PR-PS-0305	Automated acquisition	The user shall be allowed to subscribe for an automated NRT data acquisition
	Accuracy (04)	
PR-PS-0401	Threshold	▪ 15 to 20 cm for high water ▪ 10 to 15 cm for low water (highest and lowest third in the levels in the time series)
PR-PS-0402	Target	▪ 1 to 5 cm

Table 1: User Requirements for Water Level as described in ULG-CGL-TS-002

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3 ALGORITHM

3.1 OVERVIEW

Hydrological information is traditionally obtained via ground-based observation systems and networks that suffer from well-known inherent problems: high cost, sparse coverage (often limited to political local/national instead of geographical/hydrological boundaries), slow dissemination of data, heterogeneous temporal coverage, destruction of the stations during floods, absence of stations in remote areas, absence of management strategy, etc.

Although originally conceived to study open ocean processes, pioneer radar altimeter satellites (Topex Poseidon then Jason-1 as well as Envisat) have proven valuable in acquiring numerous useful measurements over inner seas, lakes, rivers and wetlands. These measurements on water level height from satellite altimeters can complement ground-based observations systems and potentially provide a major improvement in the field of continental hydrology, due to the global coverage (limited to Earth's portion at nadir of the orbital ground tracks), regular temporal sampling (Jason-3 and Sentinel-6A have a repeat cycle of 10 days, Sentinel-3A and Sentinel-3B have a cycle of 27 days) and short delivery delays (the water level products are available within 1.5 days after data acquisition).

Early studies rapidly demonstrated the great potential of satellite altimetry to monitor the water level over large continental water bodies such as the Caspian Sea, the East African lakes (Birkett, 1995; Birkett at al, 1998; Mercier et al., 2002) or the Amazon River (De Oliveira et al, 2001). However, these studies also showed the radar echoes returned by inland waters can strongly differ from those returned by the ocean, because of the contamination of the signal by reflections over emerged lands, thus potentially altering the accuracy of the height retrievals. This emphasized the need for dedicated selection criteria of the measurements to identify those associated to water targets, as well as specific processing of the radar echoes (waveforms retracking) acquired over continental waters.

However, the recognition, by the hydrological community, of the quality of altimetry measurements over inland waters is increasing. New altimeter sensors are currently being designed following the requirements expressed by hydrologists for scientific use but also for water resource management operational services that are emerging.

3.2 THE RETRIEVAL METHODOLOGY

3.2.1 Background

3.2.1.1 Satellite Radar Altimetry

Radar altimetry from space consists of vertical range measurements between the satellite and water level. The difference between the satellite altitude above a reference surface (usually a conventional ellipsoid), determined through precise orbit computation, and satellite-water surface distance, provides measurements of the water level above the reference (Figure 1).

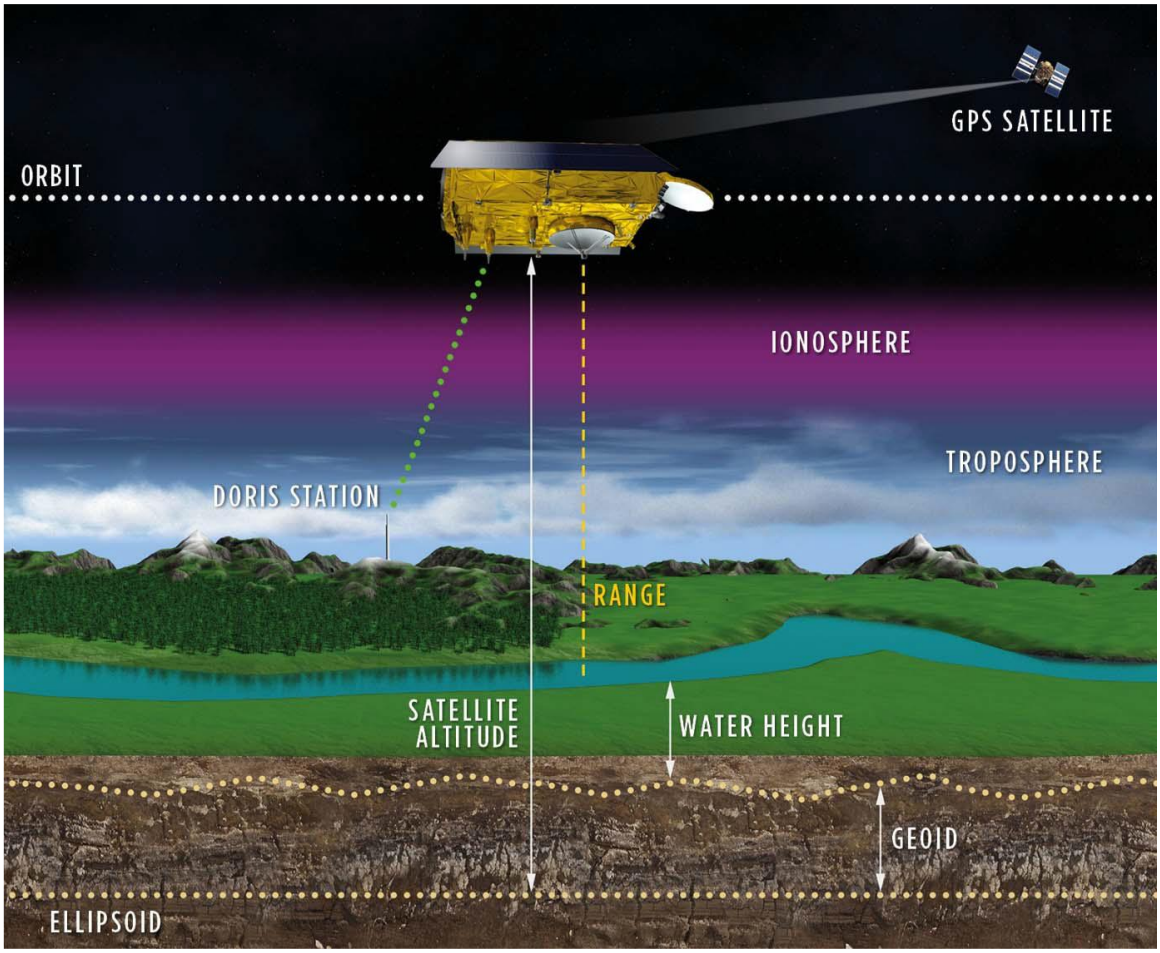


Figure 1: Altimetry principle (Credit Aviso, CNES)

The radar altimeter emits pulses towards the Earth's surface. The time which elapses from the transmission of a pulse to the reception of its echo reflected off the Earth's surface is proportional to the satellite's altitude.

Ku band (13.6 GHz) is the most frequently used frequency (used for Jason-3, Sentinel-6A, Sentinel-3A&B). It is the best compromise between the capabilities of the technology (relating to power emitted), the available bandwidth (determined by international regulations for specific applications), sensitivity to atmospheric and ionospheric perturbations. Perturbations due to water vapor in the troposphere and ionospheric electrons are accounted for by model estimates of the delay they induce on the propagation of the radar wave. These corrections are on the order of a few centimeters.

The magnitude and shape of the echoes (waveforms) also depend on the characteristics of the reflecting surface (Figure 2). The range (satellite-surface distance) is proportional to the time the radar pulse took to travel both ways from the satellite to the surface and is deduced from the position of the leading edge (the rising slope of the waveform as seen in Figure 2). Other parameters such as significant wave height can be deduced from the waveform shape. Over small lakes and rivers,

the waveform shape is very peaky, while on larger lakes, whose surfaces are affected by waves, the waveforms have a more 'oceanic' shape.

Over continental areas, the altimeter footprint frequently contains multiple surface types and not only water bodies. This contamination complexifies the waveform, therefore dedicated retrackers (post processing algorithms fitting the waveform) are used in the ground segment processing chains.

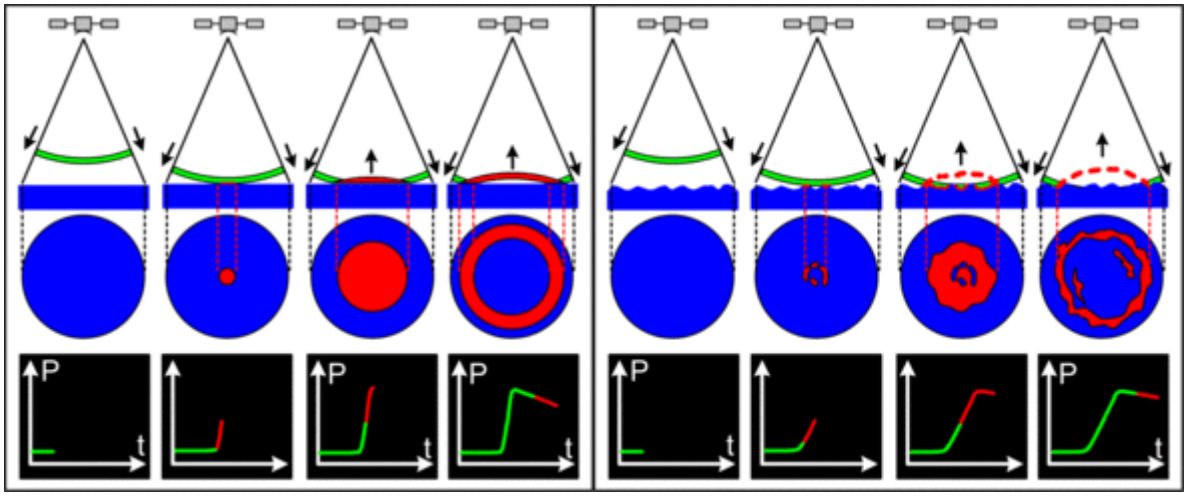


Figure 2: The radar altimeter receives the reflected echo, which varies in intensity over time. When the surface is flat (left), the reflected echo amplitude increases sharply from the moment the radar signal hits the surface. Over rough surfaces (right), the echo strikes the crest of one wave and then a series of crests which cause the reflected echo amplitude to increase more gradually. The slope of the curve representing the reflected power amplitude over time is proportional to wave height. (Credits CNES)

Water level measurements by satellite altimetry have been developed and optimized for open oceans. Nevertheless, the technique is now applied to obtain water levels of inland seas, lakes, rivers, floodplains and wetlands. Several satellite altimetry missions have been launched since the early 1990s : ERS-1 (1991-1996), TOPEX/Poseidon (1992-2006), ERS-2 (1995-2005), GFO (2000-2008), Jason-1 (2001-2012), Envisat (2002-2012), Jason-2 (2008-2019), Cryosat-2 (2010-), HY-2A (2011-), SARAL (2013-), Jason-3 (2016-), Sentinel-3A (2016-), Sentinel-3B (2018-), HY-2B (2018-), Sentinel-6A (2020-) as presented in Figure 3.

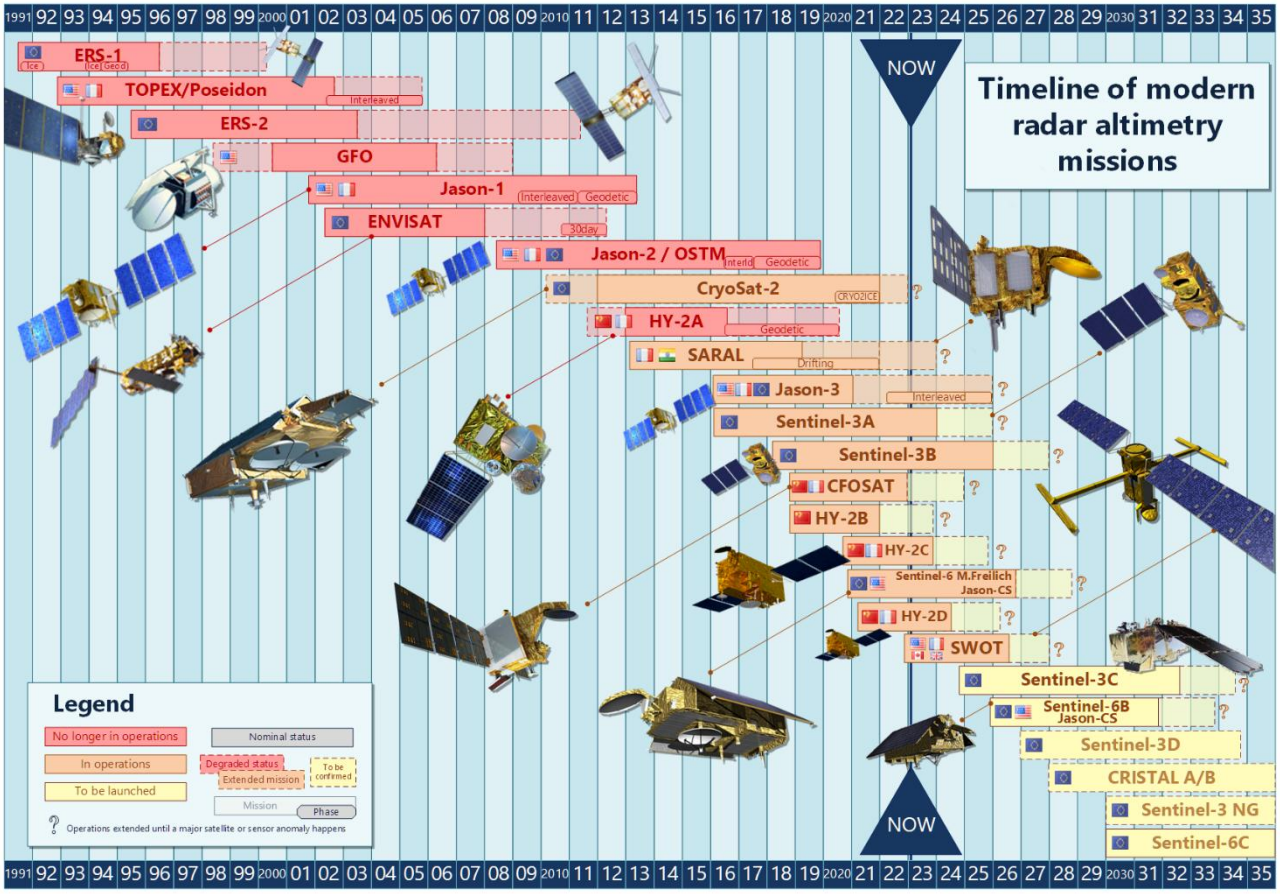


Figure 3: Aviso+ (2022). Timeline of modern radar altimetry missions. https://doi.org/10.24400/527896/A02-2022.001 version 2023/01

Recent missions use the technique of SAR altimetry that enhance performances over inland waters. As explained on the AVISO (https://www.aviso.altimetry.fr/) website, SAR-altimetry (or delay-Doppler altimetry) is a recent technique to process altimetry data. It was first operationally used for the CryoSat-2 mission. In a nutshell, the technique uses the same concept as Synthetic Aperture Radar imagers by exploiting the Doppler effect caused by the satellite movement in the along-track direction. This provides a strong spatial resolution improvement in that direction, while across-track resolution remains the same as in conventional altimetry. This is illustrated in Figure 4.

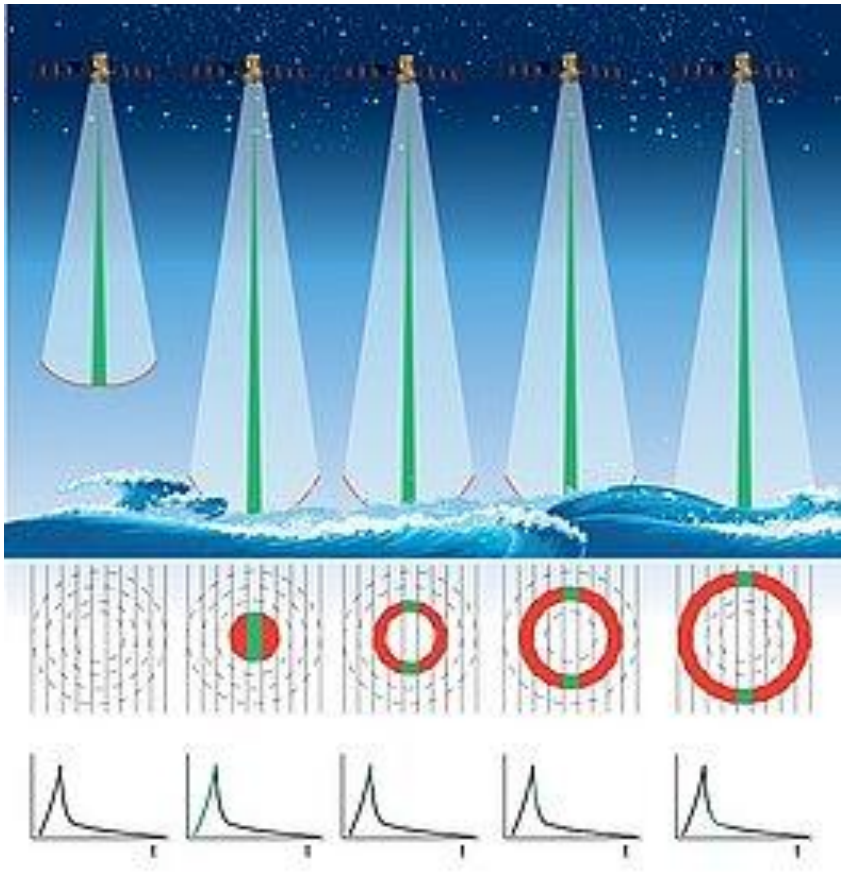


Figure 4: Formation of an echo by Doppler altimetry. The red areas represent the surfaces probe by conventional altimetry, while the green ones represent the surface probe by SAR altimetry. (Credits: AVISO, CNES/CLS/Mira Production)

ERS-1, ERS-2, Envisat and SARAL have a 35-day temporal resolution (duration of the orbital cycle) and 80 km inter-track spacing at the equator. TOPEX/Poseidon, Jason-1, Jason-2, Jason-3 and Sentinel-6A have a 10-day ground track cycle and 350 km equatorial inter-track spacing, Sentinel-3A and Sentinel-3B have a 27-day ground track cycle and 104 km equatorial inter-track spacing (Figure 5). The combined global altimetry data set has more than 20 year-long history and is intended to be continuously updated in the coming decade. Combining altimetry data from several in-orbit altimetry missions increases the spatio-temporal resolution of the sensed hydrological variables.

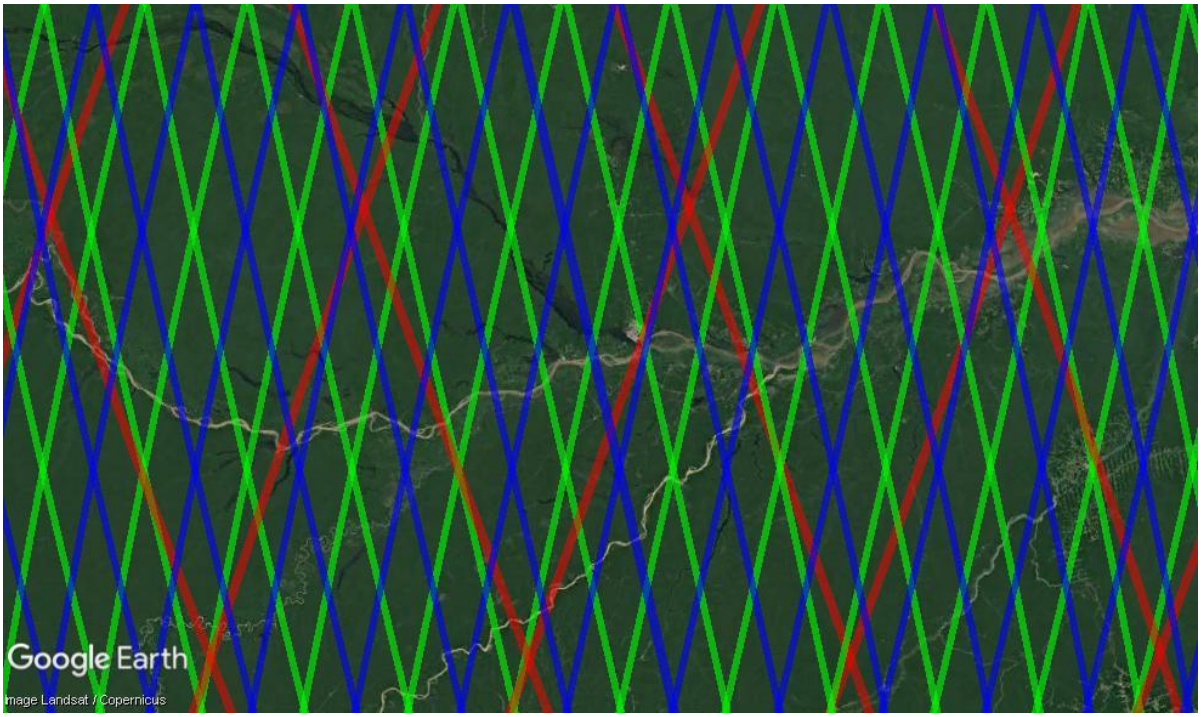


Figure 5: Jason-3/Sentinel-6A (red), Sentinel-3A (blue) and Sentinel-3B (green) pass over the Amazon. The satellites overfly each pass once every cycle (every 10 days for Jason-3/Sentinel-6, 27 days for S3A&B). Note that between those passes, no measurements are made by these satellites.

To go further in the understanding of the altimetry use for hydrology: How to use altimetry for hydrology?

3.2.1.2 Vocabulary

'Orbit' is one revolution around the Earth by the satellite.

A satellite **'Pass'** is half a revolution of the Earth by the satellite from one extreme latitude to the opposite extreme latitude.

'Repeat Cycle' is the time period that elapses until the satellite flies over the same location again.

'Range' is the satellite-to-surface pseudo distance given from the 2 ways travel time of the radar pulse from the satellite to the reflecting water body

'Retracking' is the algorithm which computes the altimetric parameters (range, backscatter coefficient...) from the radar echoes

'Altitude' is considered as the height over the reference ellipsoid

'Mean Profile' is an average over several years of the surface heights with respect to geoid, computed following the satellite's repetitive tracks.

3.2.1.3 Computing surface height

For all satellites, the following operation is done:

'Surface Height = Altitude – Corrected_range'

With:

Corrected_range = Range + Wet_tropo + Dry_ tropo + Iono + Polar tide + Solid Earth Tide

"Wet_tropo" is a correction to account for the delay of the radar signal due to cloud liquid water and water vapour in the atmosphere
"Dry_tropo" is a correction to account for the delay of the radar signal due to the dry gas component of the atmosphere.
"Iono" is a correction to account for the delay of the radar signal due the electron content of the ionosphere.
"Polar Tide" and "Solid Earth Tide" are two corrections that account for the temporal deformation of the Earth's surface.
All these corrections are provided within the input data (L2 products).

Finally, the water surface height is expressed with respect to the geoid:

Water Surface Height = Surface Height – geoid

The geoid value considered here is extracted from a mean profile file over large lakes or is computed from a geoid model in the case of rivers.

3.2.2 Processing

3.2.2.1 Lake Water Level products

Altimetry missions used are repetitive, i.e., the satellite overflies the same point at a given time interval (10 days for the Jason/Sentinel-6 satellites, 27 days for the Sentinel-3 satellites). The satellite usually does not deviate from more than +/- 1 km across its track.

A given lake can be over overflowed by several satellites, with potentially several ground tracks, depending on its surface area. In this case, each valid transect (lake crossing) is used to compute the water level time series, hence time series over lakes contain more than a point per satellite cycle.

243 lakes are operationally monitored. Several passes can be used to derive the lakes' WSH time series according to their surface areas as a lake can be overflown by several passes (see Table 2). The term "lakes" refers to lakes or reservoirs.

Mission coverage	S3A+S3B+S6A	S3A+S3B	S3A+S6A	S3B+S6A	S3A	S3B	S6A
Number of lakes	3	33	50	0	81	49	27

Table 2: Number of lakes covered by different combinations of the satellite passes.

3.2.2.1.1 Input data

The lake level measurements in the time series for periods prior to Jason-3/Sentinel-6A and S3A&B are based on merged TOPEX/Poseidon, Jason-1, Jason-2, Envisat, SARAL/AltiKa, IceSat and GFO data provided by ESA, NASA and CNES data centers. Operational updates include Sentinel-6A and Sentinel-3A&B respectively provided by CNES data center and Copernicus Hub. The chain requires the following input data:

- Altimetry data consist in Short Time Critical (STC) and Non Time Critical (NTC) data, they are respectively available about 2/3 days and 1 month after measurements acquisition. We advise the user that NTC and STC nomenclature are used for S3 and S6 data, these denominations correspond respectively to GDR and IGDR for Jason3. These 2 types of files both include the altimetric measurements as well as the necessary corrections for their exploitation (orbit, atmospheric corrections…). NTC/GDR data were used for time series initialization, but STC/IGDR data are now used for operational processing since the purpose of the Copernicus service is to provide operational water level estimates within 3 days after data acquisition. The main differences between GDR and IGDR data reside in the orbit and radiometer derived values (GDR benefit from more precise orbit values as well as more recent radiometer calibrations). Differences between these corrections are of the order of the cm.
- Mean along-track profile over the lakes
- Location information, including missions and track number for all the processed lakes

3.2.2.1.2 Method

Agencies provide altimeter L2 products in 20 Hz and 1 Hz formats (number of measurements provided per second, the 1 Hz files being created by averaging over the 20Hz measurements). The altimeter range measurements used for lakes consist of 1 Hz IGDR and GDR data for the largest lakes (Argentino, Caspian, Kara Bogaz Gol, Kariba, Lagoa do Patos, Malawi, Michigan, Nicaragua, Ontario, Rukwa, Superior, Tana, Tanganika, Tharthar, Vanerm, Victoria and Zeyskoye) and 20 Hz data for smaller ones. All classical corrections (orbit, ionospheric and tropospheric corrections, polar and solid Earth tides) are applied. Depending on the size of the lake, the satellite data may be averaged over very long distances. It is thus necessary to correct for the slope of the geoid (or equivalently, the mean lake level). Because the reference geoid provided with the altimetry

measurements (e.g., EGM96 for T/P data or EGM2008 for Sentinel-3A&B) may not be accurate enough, we have computed a mean lake level, averaging over time the altimetry measurements themselves. Such mean lake level surface along each satellite track across the lake provides a better estimate of the model geoid. Mean profiles are therefore used for lake water level computations which are then referenced with respect to this estimate of the geoid. If different satellites cover the same lake, the lake level is computed in a 3-step process. Each satellite data are processed independently. Potential radar instrument biases between different satellites are removed using T/P data as reference (see ATBD).

3.2.2.1.2.1 First step: extraction of input data

- Input data (either IGDR or GDR) are read,
- using lake files (names, pass numbers, satellite names, bias to apply, retracking model used, pass segments over the lakes) the data over each given lake are extracted.

For Sentinel-6 and Sentinel-3, (I)GDR provides high-resolution (20Hz) data for the altimetric range and altitude as well as 1-Hz averaged data. The corrections are provided at 1Hz, which is sufficient. For the biggest lakes, the data used are preferably the 1-Hz data with the range derived from ocean retracking. This retracking method is suited to fit the Brownian waveform shape induced by waves over lakes of several kilometers, as for the oceans. As for the small lakes, the high-resolution data and the ICE1 retracking are used. On small water targets the wind effect is much lower and doesn't create significant waves, hence the waveform is peakier and better fitted by the so called ICE1 retracker.

An editing is then applied to remove erroneous measurements. It consists in using thresholds on the values of the backscattering coefficient and altimetric corrections as well as validity flags provided in the L2 input products.

3.2.2.1.2.2 Second step: lake level computation and filtering

Once the relevant data are extracted and edited, mean profiles over the given passes are selected, to compute the surface height (see section 3.2.1.3). The mean profile is subtracted from the surface height computed in step 1.

Standard deviation and median values are computed. If a value is over 1.5 times the median, it is removed. Standard deviation and median values are re-computed, and measurements are removed when over twice the standard deviation. Four such iterations are done to remove abnormal points ("outliers").

3.2.2.1.2.3 Third step: output data

The final operation is the formatting of the data in GeoJSON format.

3.2.2.2 River products

Radar echoes over land surfaces are hampered by interfering reflections due to water, vegetation and topography. Consequently, waveforms (e.g., the power distribution of the radar echo within the range window) may not have the simple broad peaked shape seen over ocean surfaces, but can be complex, multi-peaked, preventing from precise determination of the altimetric height. Even when topography is smooth, problems may arise from the retracking of complex waveforms, that present several peaks due to the presence of vegetation canopy, water from wetlands, floodplains, and tributaries that are situated around the main river. In other cases, elevated topography sometimes prevents the altimeter from locking on the water surface, leading to a lower number of valid data than over smoother areas. The time series available in Copernicus Global Land are constructed over targets that were selected as good reflectors of the radar signal, meaning that they are not covered by vegetation and their water level estimate from altimetry were not detected as contaminated by echoes from surrounding other water bodies. The series are based on Sentinel-6A (since April 2022, Jason-3 beforehand) and Sentinel-3A&B I-GDRs. The basic data used for rivers are the 20 Hz data ("high rate" data).

Virtual Stations correspond to the intersection of the satellite theoretical ground track with the river (see Figure 6). Virtual Stations are defined to provide river water level time series. Virtual Stations are mono mission, their position is fixed and their time series will contain one data point per satellite cycle. To select the measurements associated to a station, a rectangular "window" (hereafter polygon file) is used to account for all available along track high-rate altimetry data over the river area, and few hundred meters around it to account for rivers smaller than 300m, over the pass of the satellite for each cycle. The coordinate of the Virtual Station is defined as the intersection of the satellite theoretical ground track and rivers centerlines. After rigorous data editing, all valid high-rate data over a transect of a given cycle are kept and their median is computed. This computation therefore concerns data separated by less than a second and acquired by a single instrument. At least two high-rate data are needed for computing the median otherwise no mean height is provided. Scattering of high-rate data with respect to the median height defines the uncertainty associated with that mean height.

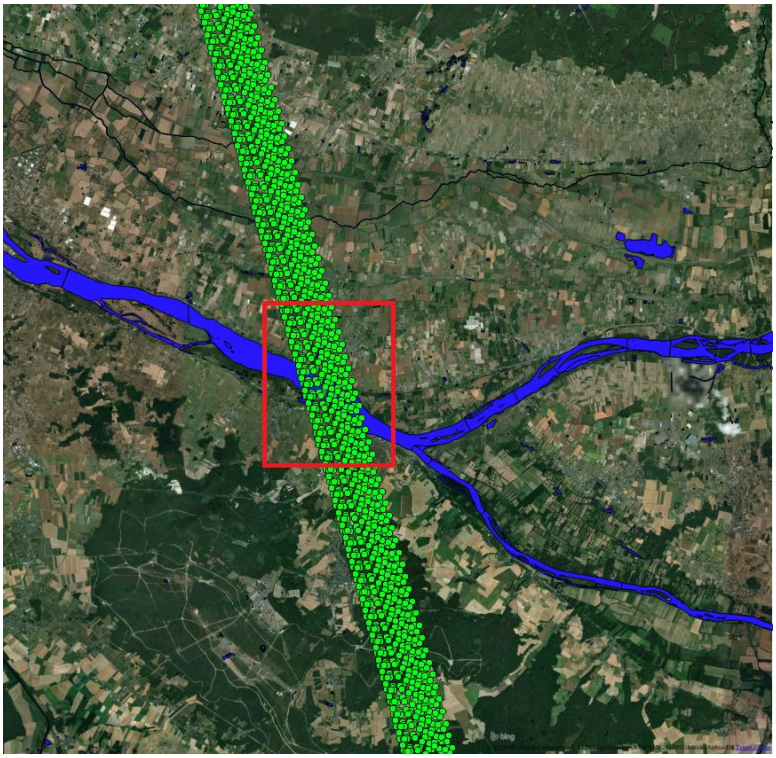


Figure 6 : Concept of definition of a Virtual Station (Sentinel-3A over the Loire River). Each point represents a 20 Hz measurement. The measurements that are aligned belong to one satellite pass over the river; those points are separated by 0.05 seconds and approximately 300m. The different 'lines' are associated with the pass of the satellite at different cycles. Each pass is separated by 27 days for Sentinel-A&B. The track slightly shifts from one pass to another (satellite control ensures this across track drift remains smaller than 1km).

The water level time series is operationally updated less than 1.5 working days after the availability of the input altimetry data for the concerned Virtual Stations.

3.2.2.2.1 Input data

Past river levels are based on TOPEX/Poseidon, Jason-2, Jason-3 L2 20Hz (10Hz for TP) data provided by ESA, NASA and CNES data centers. Operational updates include Sentinel-6A, Sentinel-3A and Sentinel-3B data provided by CNES and Copernicus Hub data centres.

- Altimetry GDR and IGDR data (IGDR for short-time critical processing), with the included corrections.
- A file with the track number to identify the L2 file to process for each Virtual Station.

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3.2.2.2.2 Method

3.2.2.2.2.1 First step: extraction and editing of data input

- Input data (either IGDR or GDR) are read,
- Using Virtual Station polygon files, the necessary data (date, location, reference surface, corrections) over each given Virtual Station are extracted.
- An editing is then applied to remove erroneous measurements. It consists in using thresholds on the values of the backscattering coefficient and altimetric corrections as well as validity flags provided in the L2 input products. This editing depends on the mission.

For all missions, high-rate (10, 20 and 40Hz depending on the mission) range and altitude data are used. The altimetric corrections used to compute the water surface height with those high-resolution data are the 1-Hz averaged ones (because not all corrections are provided at high rate), interpolated at the dates of the high-rate data. Only repetitive orbits are used so that the station is overflown at regular intervals by the altimeter, hence the data acquired during the drifting phases are not used.

3.2.2.2.2.2 Second step: data selection

The polygon file for each Virtual Station is used for data selection: an algorithm is then applied to decide if a measurement is (or is not) within the polygon.

3.2.2.2.2.3 Third step: validation and output data

The third and final step is to compute the mean, validate and produce the output data.

The mean water level processing includes:

- outlier rejection. An estimation of the mean as well as standard deviation of the water surface height time series are provided for each station in a parameter file. High-rate measurements (20Hz for Sentinel-6 and Sentinel-3A&B) with values outside mean +/- 3 sigma are identified as outliers. A sigma0 (backscatter coefficient) based criteria is also implemented to only consider data points associated with water.
- Climatology filtering: This is based on rejecting the high-rate water surface height estimates with values outside of a climatological corridor computed from the existing points of the time series (Figure 7, middle panel), see ATBD for details.
- The mean of the accepted high-rate measurements is computed over the transect to provide one value of water surface height. This value is the one provided in the time series. The associated standard deviation of the HR points at each time step is also computed and provided: it is an estimate of the uncertainty (with limitations when few high-rate measurements are available, in other words when the river is narrow).
- WGS84 ellipsoid correction: all altimetry satellites orbits are not referenced with respect to the same ellipsoid (e.g., Jason missions are referenced with respect to TOPEX/Poseidon ellipsoid while Envisat, SARAL and Sentinel orbits are referenced to WGS84). This step is meant to use the WGS84 ellipsoid as reference for all Virtual Stations water level time series.

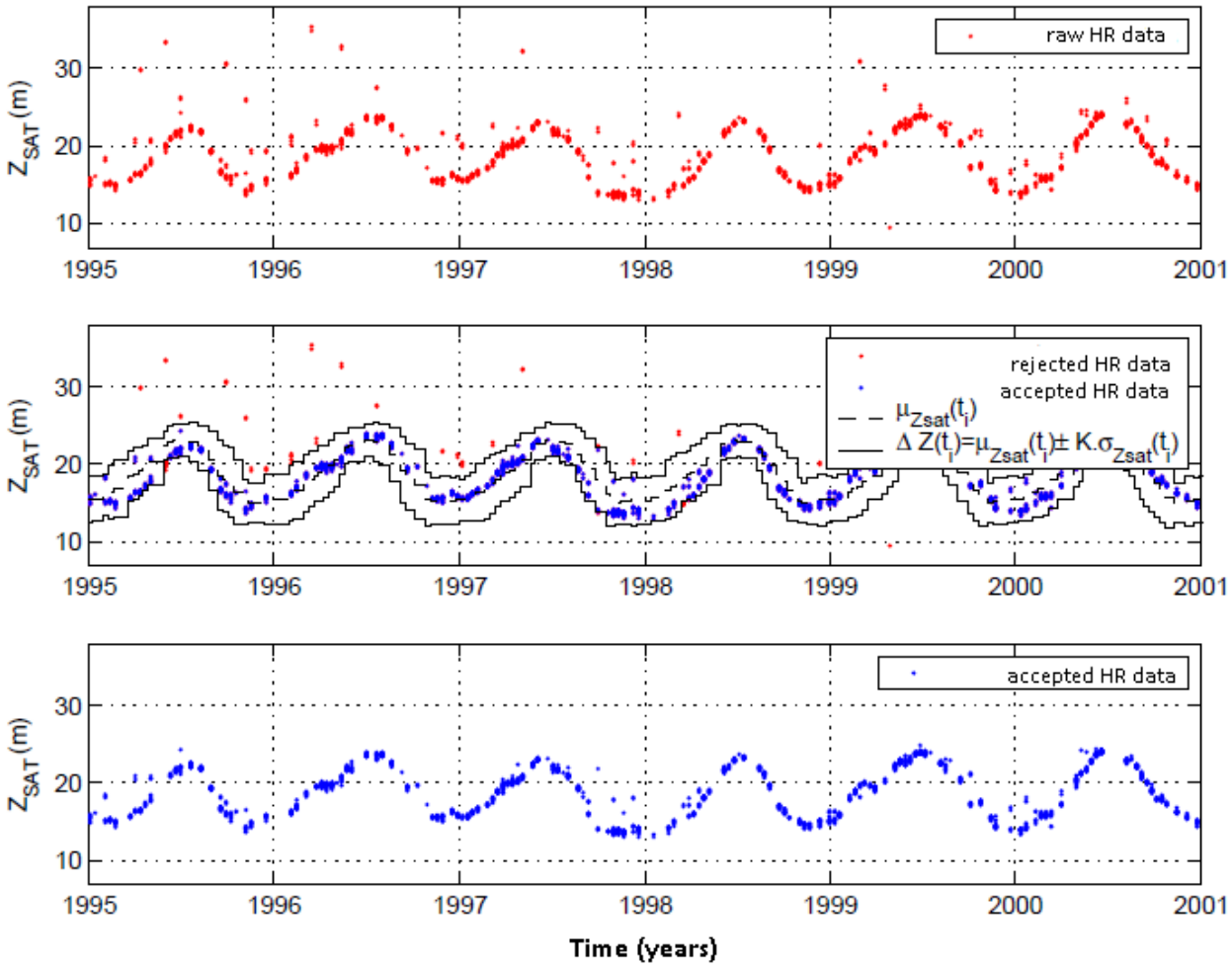


Figure 7 : Yearly filtering from Bercher 2008. Top panel: raw high rate (HR) measurements. Middle panel: application of the selection climatological corridor (details in ATBD). Bottom panel: accepted HR measurements from which the mean value is computed at each time step.

3.3 LIMITATIONS OF THE PRODUCT

The lake and river processing chains, initially developed at LEGOS, are based on the use of altimetry measurements from the CTOH database (Centre for Topography studies of the Ocean and Hydrosphere, LEGOS), including for Envisat GDRs. This CTOH expert database includes in addition some enhanced corrections that are not present in the operational L2 data sets provided by institutional suppliers (AVISO, ESA), such as for the wet and dry tropospheric as well as for the ionospheric corrections. As the operational version of HydroWeb/HYSOPE, that serves Copernicus Global Land, is using the operational L2 input data from the agencies, the altimetric corrections used by default in the processing chain are the model values included in the operational input L2 products. On the other hand, a module for calculating the height of the geoid has been developed in the operational chain, starting from a grid provided by the LEGOS.

3.4 DIFFERENCES WITH THE PREVIOUS VERSION

3.4.1 V2.2.1 to V2.2.2

Version 2.2.2 introduces few updates to the LRLW product format.

Data changes:

- Added cycle_number information
- Renamed associated uncertainty to water_surface_height_uncertainty (for rivers)
- Renamed orthometric_height_of_water_surface_at_reference_position to water_surface_height_above_reference_datum (for rivers)

Metadata changes:

- Added water_surface_reference_datum_altitude to specify the geoid level (for rivers)
- Updated processing level from LEVEL3B to LEVEL3

For historical products, newly introduced fields are populated with default placeholder values and will be corrected during future reprocessing campaigns. Example default values are: "satellite": "NA", "ground-track_number": 9999.999, "cycle_number": 9999.999

3.4.2 V2.1 to V2.2

The evolution from v2.1 to v2.2 consisted in integrating the Sentinel-6A Michael Freilich mission in the processing chain, to ensure continuity of the observations on the historical Topex/Jason ground track since April 2022 after Jason-3 was moved from this orbit. New Virtual Stations over rivers in the dataset were also included. They consist of Virtual Stations positioned at the intersection of S3A, S3B and S6A theoretical ground tracks with the rivers. We focus on including the stations for which the Open Loop Tracking Command onboard the satellites is tuned for the altimeters receiving window to be able to track the backscattered energy from these rivers. More than 11000 were added, increasing the total number of Virtual Stations operationally monitored to more than 22 000 worldwide (22 829).

3.4.3 V2 to V2.1

The evolution from v2.0 to v2.1 consisted in integrating the Sentinel-3B mission in operational mode. Stations associated with this mission and with the already used Sentinel-3A and Jason-3 mission were defined at positions between those and the satellites theoretical ground tracks. Only the high-quality time series were selected for operations. This results in the addition of more than 8000 Virtual Stations, increasing the total number of operated VS to more than 10 000. 30 new lakes are also included in this release.

3.4.4 V1 to V2

V2 includes the Sentinel-3A mission in operational mode whereas V1 only included Jason-3. The number of Jason-3 stations was also increased. Refined centrelines were computed, and new stations defined at positions between those and the satellites theoretical ground tracks. This results in the addition of more than 1000 Virtual Stations and the increase of lakes time resolution. No additional lakes have been implemented yet, waiting for feedback on the product quality and stability.

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4 PRODUCT DESCRIPTION

4.1 FILE NAMING

4.1.1 Lake products

Files are named: c_gls_WL_<*date*>_<*ID*>_ALTI_v<*x.x.x*>. json

Where

<***ID***> is the identifier of the lake (13 digits integer)

<date> the date in format yyyymmddhhmm of the last measurement in the time series

<*x.x.x*> is the product version number. The first digit refers to major version of the products. The second digit refers to minor version changes. The third one is implemented in case of reprocessing.

4.1.2 River products

Virtual Stations file names have evolved since v2.1. They are named:

c_gls_WL_<*date*>_<*ID*>_ALTI_v<*x.x.x*>. json

Where:

<**date**> is the date of the last measurement in the time series in format yyyymmddhhmm

<ID> is the Virtual Station identifier. It consists of a 13 digits integer.

<*x.x.x*> is the product version number.

The "_" delimiters remain the same as in previous versions. Only the ID identifier has evolved.

Prior to v2.1 the identifier consisted of sub expressions as follows: *ID=basinshortname-rivershortname-missionshortname-track-numberontrack*

<***basinshortname***> was a three-character code name for the hydrological basin (from the name in English, using Global Runoff Data Center database as a reference)

<***rivershortname***> was a three-character code name for the river (it can be identical to the basin code name); local names are preferred wherever possible (a choice might be done if the river is shared by several different countries)

<***missionshortname***> was the short name of the satellite mission (3 characters), among:

Code	mission full name
JA3	Jason-3
S3A	Sentinel-3A
S3B	Sentinel-3B
S6A	Sentinel-6A

<***track***>: was the track (half orbit) number; depends on the mission

<***numberoftrack***>: was the number of the Virtual Station along the given track, on 2 digits. It is useful when a river is intersected by the same satellite track at several different positions.

This previous naming relied on the accuracy of metadata, in particular river names, whose spelling can vary among the different languages, or could even be affected in the case that the river's name in the metadata had to be corrected. The new nomenclature with a unique and lifetime fixed numerical identifier solves this issue.

4.2 FILE FORMAT

The actual format is GeoJSON.

4.3 PRODUCT CONTENT

Files include:

- A tag property containing the metadata of the product
- A tag geometry containing the location of the measurements
- A tag data containing the data object (see 4.3.1.1.3) for each requested measured date

4.3.1 Data File

4.3.1.1 Lake products

4.3.1.1.1 Tag "properties" with descriptive metadata

This tag contains all the descriptive metadata including:

archive_facility	institution responsible for product archive
basin	hydrological basin or catchment name (full name), in English
comment	specific additional characteristics of the basin, lake, and product
contacts	helpdesk, accountable and owner contacts
copyright	Copernicus copyright
country	names (full name) of the countries where the lake is located (several names possible, separated by &)
credit	credit for present product
gcmd_keywords	NASA GCMD keywords relative to the present product

gemet_keywords	GEMET (GEneral Multilingual Environmental Thesaurus) keywords relative to the present product
inspire_theme	INSPIRE keywords (https://inspire.ec.europa.eu/) relative to the present product
institution	institutions responsible for the creation of this product
iso19115_topic_categories	ISO 19115 keywords relative to the present product
lake	name of the lake (full name), in local language wherever possible (a choice might be done if the shores of the lake extend over the territory of several countries)
long_name	name of the present product
missing_value	fill value in case of missing value in the present product
processing_level	data processing level
processing_mode	data processing mode
purpose	purpose of the present product
references	URL of the present product's documentation
resource	Integer identifying the lake
resource_mimetype	resource MIME type (format)
surface_of_reference	Reference used for reference area estimation
source	description of the technology used to derive the product
status	*(operational)* operational data are produced daily. At a given location, data will be updated if measurements are available (track of a satellite over the area), not updated if no measurement over the area or all measurements filtered
time_coverage_end	date of the last available measurement
time_coverage_start	date of the first available measurement
updated	date of the file last update

4.3.1.1.2 Tag "geometry"

Contains geometry as handled by the GeoJSON format. In this case, the geometry will always be of type "Point" with coordinates corresponding to the location of the center of the lake or station: [longitude, latitude]

4.3.1.1.3 Data

The data is provided as objects for each time step. The object contains 8 tags, among which two (in italic) are optional and only provided for Sentinel3A, Sentinel3B, Sentinel6A

datetime	Measure date in yyyy/mm/dd hh:mm format
identifier	Identifier of the lake
time	measure date in decimal year
water_surface_height_above_reference_datum	water surface height above surface of reference in meters
water_surface_height_uncertainty	standard deviation from height of high frequency measurements used in the estimation
satellite	*Name of the satellite input data was provided from*
ground-track_number	*Integer describing the number of the ground track (1 to 254 for Sentinel-6 and 1 to 770 for Sentinel-3)*
cycle_number	*Integer describing the number of the cycle*

4.3.1.2 River products

Tag "properties" with descriptive metadata

This tag contains all the descriptive metadata including:

archive_facility	institution responsible for product archive
basin	hydrological basin or catchment name (full name), in English
comment	specific additional characteristics of the basin, river, and product

contacts	helpdesk, accountable and owner contacts
copyright	Copernicus copyright
country	names (full name) of the countries where the river is located (several names possible, separated by &)
credit	credit for present product
gcmd_keywords	NASA GCMD keywords relative to the present product
gemet_keywords	GEMET (GEneral Multilingual Environmental Thesaurus) keywords relative to the present product
inspire_theme	INSPIRE keywords (https://inspire.ec.europa.eu/) relative to the present product
institution	institutions responsible for the creation of this product
iso19115_topic_categories	ISO 19115 keywords relative to the present product
long_name	name of the present product
missing_value	fill value in case of missing value in the present product
platform	name of the altimetry satellite platform used in the present product
processing_level	data processing level
processing_mode	data processing mode
purpose	purpose of the present product
references	URL of the present product's documentation
resource	Integer identifying the virtual station
resource_mimetype	resource MIME type (format)
river	name of the river (full name), in local language wherever possible (a choice might be done if the river extends over the territory of several different countries)
source	description of the technology used to derive the product, "Derived from satellite altimetry"
status	*(operational)* operational data are produced daily. At a given location, data will be updated if measurements are available (track of a satellite over the area), not

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	updated if there are no measurements over the area or all measurements are filtered
time_coverage_end	date of the last available measurement
time_coverage_start	date of the first available measurement
updated	date of the file last update
water_surface_reference_datum_altitude	altitude of the water surface reference (geoid) model in meter
water_surface_reference_name	name of the water surface reference (geoid) model

4.3.1.2.1 Tag "geometry"

Contains geometry as handled by the GeoJSON format. In this case, the geometry will always be of type "Point" with coordinates corresponding to the location of the Virtual Station: [longitude, latitude]

4.3.1.2.2 Data

The data is provided as objects for each time step. The object contains 8 tags:

identifier	Identifier of the virtual station
time	measure date in decimal year
datetime	Measure date in yyyy/mm/dd hh:mm format
water_surface_height_above_reference_datum	water surface height above surface of reference in meters
water_surface_height_uncertainty	standard deviation from height of high frequency measurements used in the estimation
satellite	Name of the satellite input data was provided from
ground-track_number	Integer describing the number of the ground track (1 to 254 for Sentinel-6 and 1 to 770 for Sentinel-3)
cycle_number	Integer describing the number of the cycle

4.3.2 Quicklook

4.3.2.1 Lakes products

The two figures below (Figure 8 and Figure 9) illustrate the content of the data file for Lake Baïkal (Russia). Note that the time series has been computed since late 1992, starting with TOPEX/Poseidon until to 2002, with Jason-1 between 2002 and 2008, Jason-2 between 2008 and 2016, Jason-3 in between October 2016 and April 2022, Sentinel-6A since April 2022, Sentinel-3A since June 2016 and Sentinel-3B since 2019.


```
{
    "type": "Feature",
    "geometry": {
        "type": "Point",
        "coordinates": [
            108.17,
            53.58
        ]
    },
    "properties": {"resource": "1300000000011",
        "basin": "Yenisei",
        "comment": "Timeseries specified by LEGOS and computed by CLS on behalf of CNES
and Copernicus Global Land",
        "lake": "baikal",
        "surface_of_reference": "GGMO2C; high resolution global gravity model",
        "country": "Russia",
        "institution": "LEGOS, CLS, CNES",
        "source": "Derived from satellite altimetry",
        "references": "http://land.copernicus.eu/global/products/wl",
        "archive_facility": "CLS",
        "time_coverage_start": "1992-09-27 20:09",
        "time_coverage_end": "2026-06-14 18:27",
        "updated": "2026-06-16 00:35",
        "long_name": "Lake and River Water Level",
        "processing_level": "LEVEL3",
        "processing_mode": "Near Real Time",
        "status": "operational",
        "copyright": "Copernicus Service Information 2017",
        "contacts": "Helpdesk: http://land.copernicus.eu/global/contactpage \n
Accountable contact: European Commission DG Joint Research Centre
copernicuslandproducts@jrc.ec.europa.eu \n Owner contact: European Commission DG Internal
Market, Industry, Entrepreneurship and SMEs ENTR-COPERNICUS-ASSETS@ec.europa.eu",
        "inspire_theme": "Hydrography",
        "gemet_keywords": "water level, water management, hydrometry, hydrology, climate,
seasonal variation, environmental data, environmental monitoring, monitoring, remote
sensing",
        "gcmd_keywords": "TERRESTRIAL HYDROSPHERE, SURFACE WATER",
        "iso19115_topic_categories": "elevation,inlandWaters",
        "credit": "Lake and River Water Level products are generated by the Global Land
Service of Copernicus, the Earth Observation Programme of the European Commission and the
Theia-land program supported by CNES. The research leading to the current version of the
product has received funding from CNES, LEGOS, IRD and CLS.",
        "purpose": "This product is first designed to fit the requirements of the Global
Land component of Land Service of Copernicus. It can be also useful for all applications
related to the environment monitoring, climate research and hydrology.",
        "resource_mimetype": "application/json",
        "missing_value": 9999.999
    },
    "data": [
```

```
        {
            "identifier": "L_baikal",
            "time": 1992.7399988616,
            "datetime": "1992/09/27 20:09",
            "water_surface_height_above_reference_datum": 455.82,
            "water_surface_height_uncertainty": 0.13,
            "satellite": "NA",
            "ground-track_number": 9999.999,
            "cycle_number": 9999.999
        },
        {
            "identifier": "L_baikal",
            "time": 1992.7899988616,
            "datetime": "1992/10/16 03:21",
            "water_surface_height_above_reference_datum": 455.77,
            "water_surface_height_uncertainty": 0.1,
            "satellite": "NA",
            "ground-track_number": 9999.999,
            "cycle_number": 9999.999
        },

[…]
        {
            "identifier": "L_baikal",
            "time": 2025.7613565449,
            "datetime": "2025/10/05 21:29",
            "water_surface_height_above_reference_datum": 455.99,
            "water_surface_height_uncertainty": 0,
            "satellite": "sentinel6A",
            "ground-track_number": 214,
            "cycle_number": 9999.999
        } [...]
    ]
}
```


Figure 8: Reproduction of the lines of the water level product for Lake Baïkal (Russia) in GeoJSON format. Note the optional "satellite', "ground-track_number" and "cycle_number" values (see section) 4.3.1.1.3

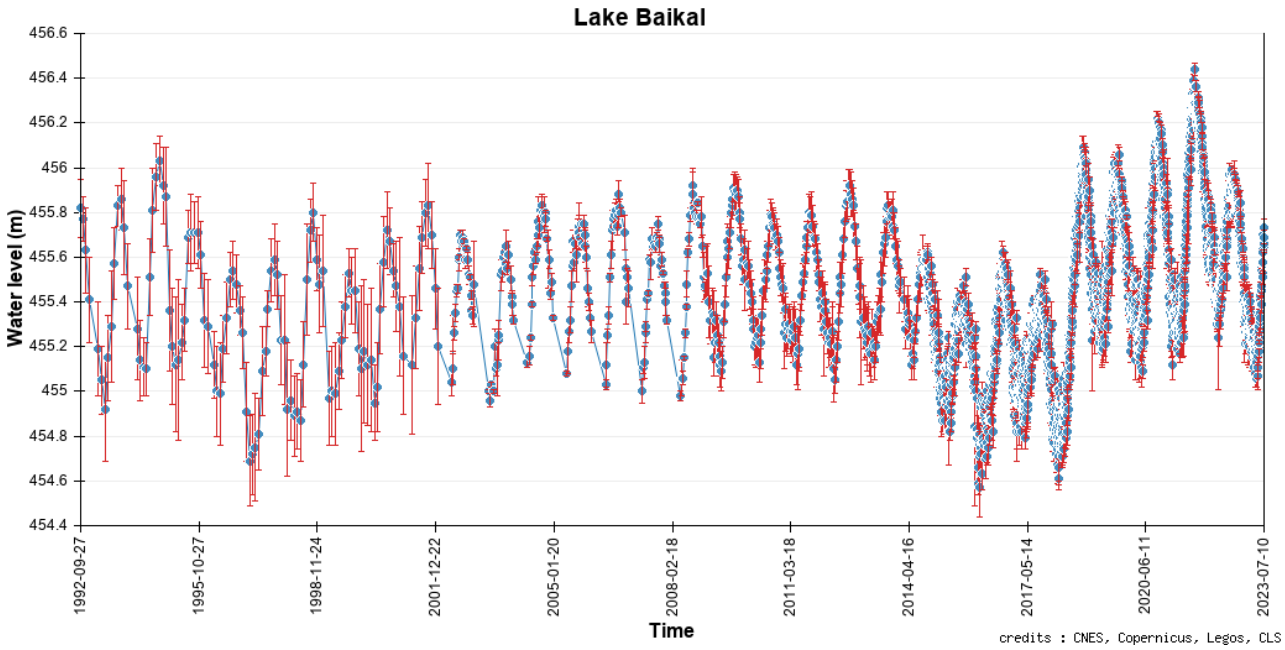


Figure 9: Plot of the water level time series (blue) and associated uncertainties (red) for Lake Baïkal (Russia).

4.3.2.2 River products

The two figures below (Figure 10 and Figure 11) show an example of the content of a data file on a Virtual Station on the Danube River along with its graphical representation.


```
{
    "type": "Feature",
    "geometry": {
        "type": "Point",
        "coordinates": [
            20.63,
            44.79
        ]
    },
    "properties": {
        "resource": "0000000004517",
        "basin": "Danube",
        "river": "Dunav",
        "water_surface_reference_datum_altitude": 44.26,
        "comment": "Timeseries specified by LEGOS and computed by CLS on behalf of CNES
and Copernicus Global Land",
        "country": "Serbia",
        "institution": "LEGOS, CLS, CNES",
        "source": "Derived from satellite altimetry",
        "references": "http://land.copernicus.eu/global/products/wl",
        "archive_facility": "CLS",
        "time_coverage_start": "2016-08-22 20:02",
        "time_coverage_end": "2026-05-26 20:03",
        "updated": "2026-05-29 00:44",
        "platform": "SENTINEL3A",
        "long_name": "Lake and River Water Level",
        "processing_level": "LEVEL3",
        "processing_mode": "Near Real Time",
```

```
        "status": "operational",
        "copyright": "Copernicus Service Information 2017",
        "contacts": "Helpdesk: http://land.copernicus.eu/global/contactpage \n
Accountable contact: European Commission DG Joint Research Centre
copernicuslandproducts@jrc.ec.europa.eu \n Owner contact: European Commission DG Internal
Market, Industry, Entrepreneurship and SMEs ENTR-COPERNICUS-ASSETS@ec.europa.eu",
        "inspire_theme": "Hydrography",
        "gemet_keywords": "water level, water management, hydrometry, hydrology, climate,
seasonal variation, environmental data, environmental monitoring, monitoring, remote
sensing",
        "gcmd_keywords": "TERRESTRIAL HYDROSPHERE, SURFACE WATER",
        "iso19115_topic_categories": "elevation,inlandWaters",
        "credit": "Lake and River Water Level products are generated by the Global Land
Service of Copernicus, the Earth Observation Programme of the European Commission and the
Theia-land program supported by CNES. The research leading to the current version of the
product has received funding from CNES, LEGOS, IRD and CLS.",
        "purpose": "This product is first designed to fit the requirements of the Global
Land component of Land Service of Copernicus. It can be also useful for all applications
related to the environment monitoring, climate research and hydrology.",
        "resource_mimetype": "application/json",
        "water_surface_reference_name": "EGM2008",
        "missing_value": 9999.999
    },
    "data": [
        {
            "identifier": "R_DANUBE_DUNAV_KM1205",
            "time": 2016.6416249241,
            "datetime": "2016/08/22 20:02",
            "water_surface_height_above_reference_datum": 69.8,
            "water_surface_height_uncertainty": 0.01,
            "satellite": "S3A",
            "ground-track_number": 27,
            "cycle_number": 8
        },
        [...]
        {
            "identifier» : "R_DANUBE_DUNAV_KM1205",
            "time": 2025.7337956621,
            "datetime": "2025/09/25 20:03",
            "water_surface_height_above_reference_datum": 68.72,
            "water_surface_height_uncertainty": 0.23,
            "satellite": "S3A",
            "ground-track_number": 27,
            "cycle_number": 131
        }
    ]
}
```


Figure 10: Reproduction of the lines of the water level product for one given Virtual Station over the Danube River, with header and part of the data lines.

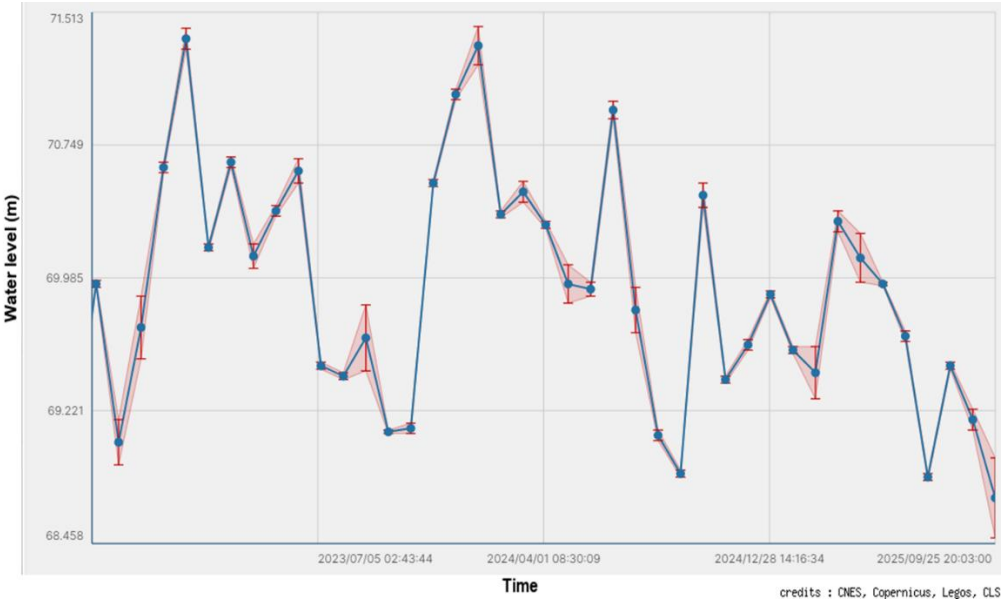


Figure 11: Example of a water level time series on Dunav river.

4.4 PRODUCT CHARACTERISTICS

4.4.1 Projection and Grid Information

Longitude and Latitude values are expressed with respect to the WGS84 ellipsoid.

Water level values are geoidal heights expressed with respect to EGM2008 model, unless stated different in the header of the product.

4.4.2 Spatial Information

As described above, the water level heights can only be measured right below the satellite's orbital ground tracks (nadir), if favorable measurement conditions are encountered (good reflected radar signal, low contamination from other echogenic sources, see §3.2.2.2 River products). It is unfortunately not always possible to retrieve valuable measurements at each crossing between hydrographic networks and satellites ground tracks.

In addition, given the inclination of the orbit of some altimetry missions, such as Jason-3/Sentinel-6A that is limited to +/- 66° in latitude, some targets in the Arctic region are not overflown.

The ground track coverage of all altimetry missions can be found on AVISO website: http://www.aviso.altimetry.fr/en/data/tools/pass-locator.html.

As an example, the offset between 2 ground tracks at the Equator is 315 km for Jason-3/Sentinel-6A tracks (10-days revisit time), 80 km for Envisat/Saral (35-days revisit time) and 104 km for Sentinel-3 (27-days revisit time).

4.4.3 Temporal Information

For Virtual Stations, the revisit time corresponds to the duration of the orbital cycle of the considered satellite: 10 days for Jason-3/Sentinel-6, 27 days for Sentinel-3A and Sentinel-3B.

Lake level time series may be updated more frequently in case there are several orbital ground tracks of a given satellite that cross the lake, or if several satellites fly over the lake.

Nominally, the water level time series, either lakes or rivers, are updated within 2 days after acquisition of measurements by the operational satellite(s).

4.4.4 Data Policies

Any use of the RiverAndLakeWaterLevel product assumes the obligation to include in any publication or communication using these products the following citation:

"The product was extracted from land service of Copernicus, the Earth Observation program of the European Commission. The research leading to the current version of the product has received funding from various European Commission Research and Technical Development programs. The product is based on THEIA HYDROWEB products specified by LEGOS and operated by CLS on behalf of CNES."

The user accepts to inform Copernicus about the outcome of the use of the above-mentioned products and to send a copy of any publication that uses these products to the support contact below.

4.4.5 Contacts

More information and documentation about the product is available from the Copernicus Global Land Service web site at https://land.copernicus.eu/global/products/wl

For support, the Help Desk service staff are the primary point of contact:

https://land.copernicus.eu/global/contact

Accountable contact:

European Commission, Directorate-General Joint Research Centre, Italy
Email address: copernicuslandproducts@jrc.ec.europa.eu

5 VALIDATION

Based on the user requirements issued in Section 2, the quality of the products may be assessed as overall satisfactory. More than 22 000 Virtual Stations are already fully operational. As detailed in the Quality Assessment Report document, the quantity and global quality of the products has improved (more targets, smaller dispersion of the measurements) thanks to the integration of the Sentinel-3A mission in the V2, then Sentinel-3B in the V2.1 of the products and Sentinel-6A in V2.2, with respect to previous V1 (deprecated, not available for diffusion anymore).

As detailed in the QAR, absolute validation metrics show that the measurements precision is of the order of 10 to 25cm, depending on the hydrological basins. The mean percentage of provided measurements in the time series relative to the number of satellite passes (completeness) per basin is higher than 95%. Some stations over the Congo basin present a lower percentage of available measurements since they were used as calibration areas for Jason-3. Correlations between stations situated on a same river were also assessed. The correlations are very good (R^2>0.8) over the Amazon and African basins. Correlations are smaller over more anthropized basins. Validation metrics with respect to a DEM or In Situ measurements confirm these results.

Lake time series quality is also assessed using both absolute (comparisons within the product) and relative (comparison to other satellite and in situ products) validation metrics. Correlation coefficients with both external altimetry datasets are very high (>0.9 most of the time). The RMS is of the order of 10-20 cm for most lakes. We nevertheless point out that absolute reference can be questioned on some lakes either in the C-GLOPS or the comparison datasets. Comparisons to In-Situ datasets present very high correlation coefficients (> 0.9) and u-RMSE with a median value of 13cm over 69 altimetric / in-situ time series comparisons.

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